

AI-Powered Harvest Yield Prediction for Non-Cash Crops (Big Onion) in Sri Lanka

Pre-harvest, district-level yield forecasting from open weather, satellite and soil data — with calibrated uncertainty.

1 Problem Statement

Sri Lanka consumes roughly **220,000 metric tons** of big onion each year. Domestic cultivation covers only part of that demand and the balance is imported at a direct foreign-exchange cost.

Unlike paddy, big onion has **no crop-cutting survey methodology**. Yield figures are compiled from the subjective assessments of agricultural officers, they carry no quantified error, and they are finalised only **after** the crop has been harvested and sold — while import volumes must be committed months in advance.



2 Project Objectives

- Data foundation** — integrate district yield, daily weather, satellite vegetation and soil data into one clean, reproducible dataset.
- Feature engineering** — derive agronomically meaningful predictors of thermal, water and canopy status.
- Comparative modelling** — train classical, deep, symbolic and physics-hybrid models under one identical protocol.
- Honest evaluation** — leakage-free cross-validation, paired significance testing and calibrated prediction intervals.
- Attribution and ablation** — quantify what each data source and each predictor actually contributes.
- Delivery** — a documented REST service and a decision-support dashboard for non-technical users.

3 Methodology

Data. Four open streams — DCS district yield, NASA POWER and CHIRPS weather, MODIS and Sentinel-2 vegetation and land-surface temperature via Google Earth Engine, and ISRIC SoilGrids — are reduced to one record per district, season and year.

Features. 32 predictors in five groups: weather aggregates (9), vegetation dynamics (11), yield history (5), soil (4) and three cross-modal interaction terms encoding agronomic hypotheses.

Models. Nine predictors trained identically — Random Forest, XGBoost, SVR, LSTM, BiLSTM, 1D-CNN, hybrid CNN-LSTM, symbolic regression and a physics-residual hybrid — plus a constrained convex stacking layer.

Evaluation. **Leave-One-Year-Out cross-validation:** an entire year is held out, hyperparameters are tuned inside the training partition only, and every fitted quantity is re-estimated per fold. A random split would leak information between districts of the same year and flatter the result. Models are compared with the paired Wilcoxon signed-rank test, and split-conformal intervals are attached to every forecast.

4 Key Modules & Components

Module	Responsibility	Owner
Data pipeline	Source ingestion, key reconciliation, season windowing, cleaning, 32-predictor engineering	Sharuja B.
Prediction engine	Nine models, LOYO-CV harness, stacking, conformal calibration, SHAP attribution	Arkam B.H.M.
REST service	Validated prediction endpoint with interval, attribution and provenance; 8 endpoints	Arkam B.H.M.
Dashboard	Choropleth overview, prediction form with context prefill, explainability and advisory views	Shathurya P.

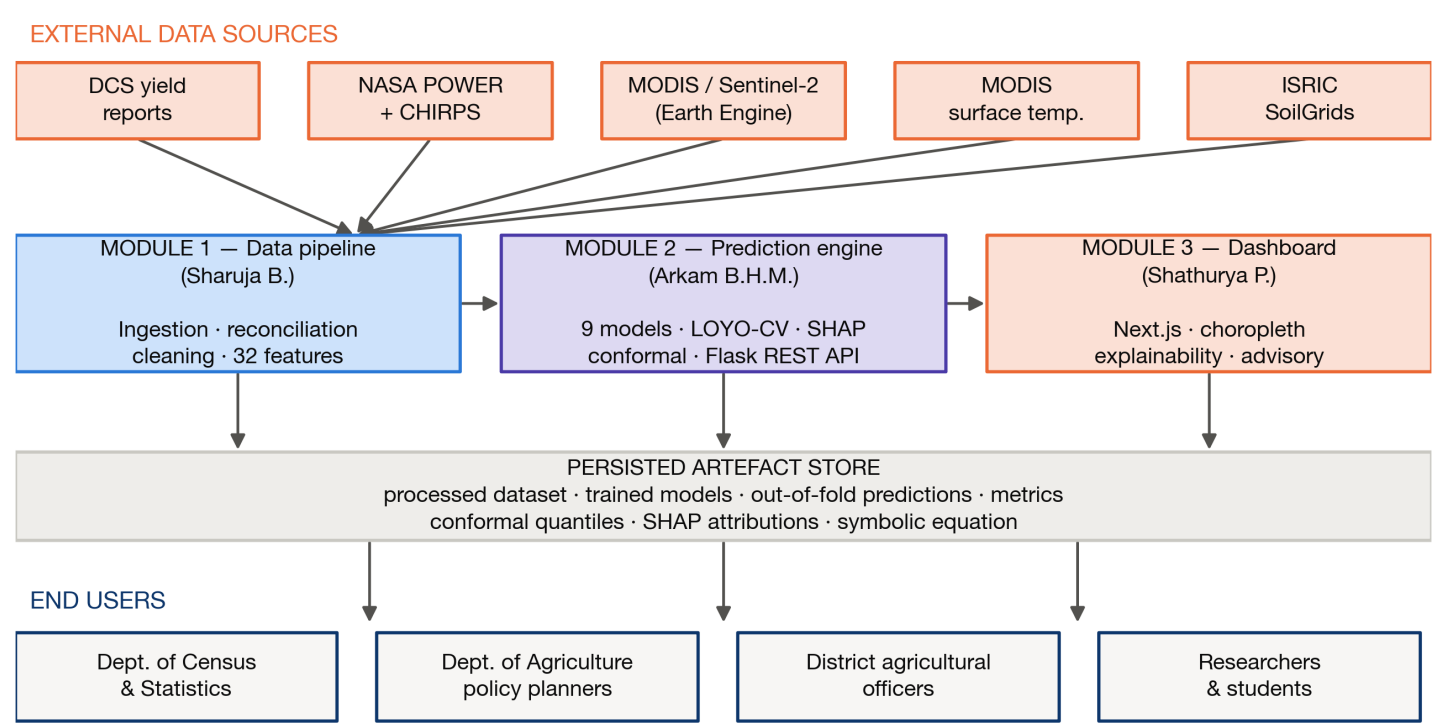
5 Research Novelty

- 1 — Physics-residual hybrid

A closed-form agronomic backbone (FAO-33 water response, growing degree days, heat stress) supplies what would otherwise have to be learned; a Random Forest models only the residual. Just two parameters are fitted, and they are refitted inside every fold.
- 2 — Season-aware hybrid CNN-LSTM

A CNN branch reads the vegetation trajectory and an LSTM branch the weather sequence. The Yala/Maha indicator is injected after feature extraction, so both branches share parameters across seasons and only the dense head learns season-specific yield baselines.

6 System Architecture



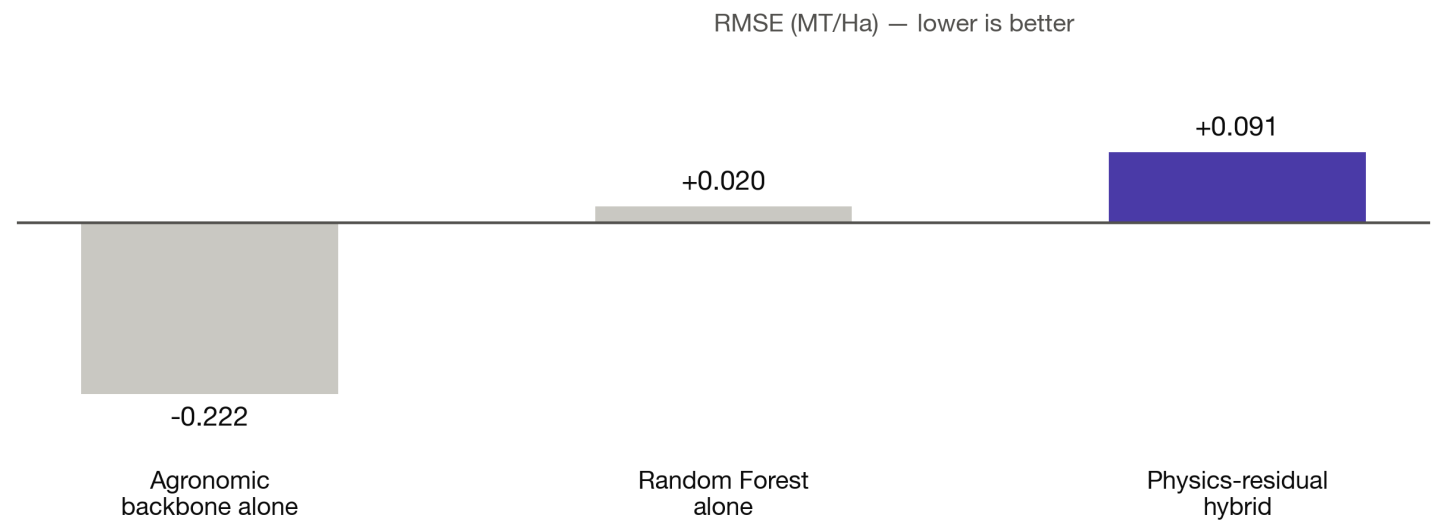
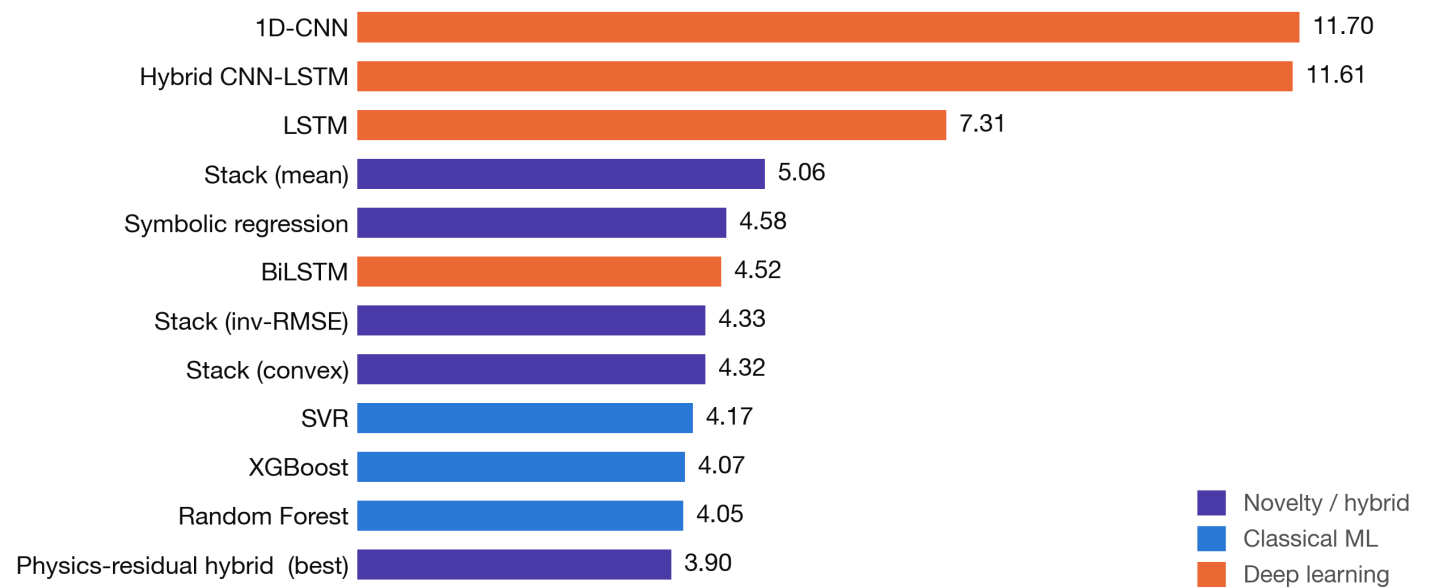
Four external data streams feed three build modules; the serving layer reads persisted artefacts and never trains.

7 Key Results

BEST MODEL — Physics-residual hybrid

RMSE 3.90 MT/Ha · MAE 3.35 MT/Ha · R² 0.091 · MAPE 23.3 % · 90 % interval ±6.85 MT/Ha

Leave-One-Year-Out cross-validation over 28 collected seasonal records (4 districts × 7 years, Yala).



The agronomic prior lifts R² by +0.070 over the same learner used alone — the project's clearest positive finding.

8 Outcomes & Conclusion

- Classical models win decisively.** Every deep architecture scores below a constant-mean predictor; the paired Wilcoxon test rejects parity at p < 0.0001. At 28 records, capacity is a liability.
- Domain knowledge beats extra parameters.** The mechanistic backbone improves R² from 0.020 to 0.091, and the convex stack independently assigns it the largest weight (0.399).
- No single data stream suffices.** A six-configuration ablation puts every individual source below the baseline; only the full set approaches it, and the two engineered interaction terms outrank every raw measurement in the SHAP ranking.
- The limit is data, not engineering.** Identical code reaches R² 0.842 on a larger reference sample, so the constraint is the size of the obtainable record.
- Next** — extend the record backwards and to the Maha season, mask satellite aggregation to cultivated onion extent, and forecast at operational lead time.

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